UMANG P. KAKLOTAR

COMPUTER VISION ENGINEER | AI & DEEP LEARNING DEVELOPER

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PROFESSIONAL SUMMARY

Computer Vision & Deep Learning Developer specializing in Automated Optical Inspection (AOI), real-time YOLOv8 object detection, and ROS 2 / Gazebo industrial digital twin simulations. Experienced in OpenCV, PyTorch, FastAPI, and Multimodal RAG systems. Proven track record building end-to-end inspection systems from custom model training to shop-floor mobile camera interfaces and database defect tracking.

TECHNICAL SKILLS

Computer Vision & DL:	Ultralytics YOLOv8, YOLO, OpenCV, PyTorch, Torchvision, Scikit-Image, Object Detection, Bounding Box Annotation, Quality Classification, Defect Localization, Transfer Learning.
Robotics & Simulation:	ROS 2 (Jazzy), Gazebo Sim 8, Sensor Cameras, Conveyor Automation, CvBridge, Virtual Proximity Sensors.
AI, Vector DB & LLMs:	Google Gemini Embedding-2, FAISS (IndexFlatIP), Groq LPU, LangChain, Multi-tenant RAG, Speech Recognition.
Languages & Backend:	Python (FastAPI, NumPy, Pandas, Scikit-Learn), JavaScript, React 18, Vite, Node.js, HTML5, CSS3, REST APIs, SSE.
Databases & Tools:	MongoDB, MySQL, Git, GitHub, Postman, Linux/Bash, VS Code, Clerk Auth.

KEY ENGINEERING PROJECTS

PCB AI Defect Inspection & Smart Manufacturing System — FLAGSHIP CV PROJECT

YOLOv8 (best.pt), OpenCV, ROS 2 Jazzy, Gazebo Sim 8, FastAPI, MongoDB, React | Live: pcb-frontend.getvoroa.com

- **Custom YOLOv8 Training:** Trained custom model identifying 7 PCB defect classes (Copper, Mousebite, Open, Pin-hole, Short, Spur, Missing Hole) with >77-92% confidence.
- **Web & Mobile Camera Pipeline:** Built dual-surface optical inspection accepting image uploads and live camera streams with instant bounding boxes, defect count, and quality status (Pass/Rework/Scrap).
- **ROS 2 / Gazebo Digital Twin:** Simulated virtual conveyor inspection line in Gazebo Sim 8 with ROS 2 nodes, virtual proximity sensors, and CvBridge camera feeds triggering real-time inference.
- **Traceability & Telemetry:** Integrated FastAPI backend with MongoDB to record PCB IDs, timestamps, defect coordinates, and machine health telemetry.

Document AI — Enterprise Multi-Format RAG Platform — FULL-STACK AI

Gemini Embedding-2, FAISS, Groq LPU, LangChain, FastAPI, Clerk, MongoDB, React | Live: rag-frontend-omega-three.vercel.app

- **Sub-Millisecond Vector Search:** Implemented semantic chunking (900/150 overlap) across PDF, Word, CSV; embedded via Gemini-2 into in-memory FAISS IndexFlatIP with sub-ms retrieval.
- **Real-Time Streaming & Citations:** Streamed answers via Server-Sent Events using Groq LPU with zero hallucination prompts and citation bubbles displaying source file and page.
- **Multi-Tenant Isolation:** Secured with Clerk JWT auth, isolating vector index spaces and MongoDB session chats per user.

Real Estate Valuation & Home Price Prediction System — APPLIED MACHINE LEARNING

- **Predictive Modeling & EDA:** Developed real estate regression model with Scikit-Learn; performed automated outlier removal, feature engineering, and cross-validation to minimize RMSE.

WORK EXPERIENCE

Frontend Developer Intern — Jemistry Info Solutions LLP — 6 MONTHS | 2025

- **Web Development:** Built responsive web interfaces converting Figma designs using HTML5, CSS3, JavaScript, React; integrated backend REST APIs with Axios, managed version control in Git/GitHub.

EDUCATION

Bachelor of Computer Applications (BCA) — Smt ZS Patel College (Surat) — 2022 - 2025

- **Core Focus:** Computer Vision & Image Processing, Machine Learning, Data Structures & Algorithms, DBMS, Web Technologies.